

From Memory to Belief: A Survey of State Maintenance and Belief Revision in LLM Decision Agents

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Abstract

Language-model agents now plan, call tools, and act over many steps, and the failures that dominate reports from such settings are failures of state rather than of reasoning: the agent loses track of what it has learned, keeps acting on evidence that has since been contradicted, or revises what it believes without changing what it does. Memory mechanisms, uncertainty estimation, and post-training with denser rewards each address one part of this problem, but the object that connects them, a persistent, uncertainty-aware belief about a partially observed world, has no survey of its own. This survey takes the belief state as its organising object and traces its development from 2020 to 2026, from context and memory stores through external Bayesian filters and model-written beliefs to learned world models exposed to the policy. We ask one question of every method, *which object receives credit*, the output, the trajectory, the interaction, or the state, and use it to organise three axes: how the belief is represented and whether it commits to testable predictions, how it is revised and where revision fails, and how learning signals reach it. We then align the fragmented belief-level evaluation literature by construct, catalogue 33 benchmarks, and identify the gap the alignment exposes: no published method yet obtains a signed learning signal for a written belief from the agent's own predictions.

1 Introduction

Language-model agents now plan, call tools, retrieve evidence, and act over horizons of hours or months, and the failures reported from these settings are failures of state rather than of reasoning. A model that solves a problem in one pass loses much of its accuracy when the same information arrives over several turns (Laban et al., 2025); an agent that plans well falls to near-random success once the environment moves while it deliberates (Gonzalez-Pumariega et al., 2025); an agent

Survey (venue)	Organising axis	B	R	S	E
MEMORY					
Zhang et al. (2024b) (arXiv)	design, evaluation, use	○	●	○	○
Hu et al. (2025b) (arXiv)	factual / experiential / working	○	●	○	●
Luo et al. (2026a) (ACL F)	storage → reflection → experience	○	●	○	○
REWARDS AND CREDIT					
Wu (2025) (EMNLP F)	reward model × stage	○	○	○	○
Zheng et al. (2025) (ACL)	data → PRM → use	○	○	○	○
Zhang et al. (2025) (TMLR)	capability × application	●	○	○	○
Zhang (2026) (arXiv)	granularity × method	○	○	●	○
UNCERTAINTY					
Xia et al. (2025) (ACL F)	four estimation avenues	○	○	○	●
Lin et al. (2026b) (arXiv)	what / how / where	●	○	○	●
EVALUATION, HORIZON, DOMAIN					
Yehudai et al. (2026) (ACL F)	capability × domain	○	○	○	○
Chen et al. (2026a) (arXiv)	lifecycle × horizon carrier	●	●	○	○
Xu et al. (2026c) (arXiv)	forecasting architecture	●	○	○	●
Dong et al. (2025b) (EMNLP F)	financial function	○	○	○	○
This survey	credited object	●	●	●	●

Table 1: Neighbouring surveys and the belief state. **B**: belief as an object of its own; **R**: revision and its failure modes; **S**: learning signals that reach the belief; **E**: belief-level evaluation. ● yes, ○ partly, ○ no.

given only observations wins a fraction of the episodes won by one that reads the simulator's latent state (Alaswad et al., 2026); and in markets, where evidence arrives late and partially, stronger backbones do not reliably produce stronger trading agents (Xie et al., 2024; Li et al., 2024). What these failures share is that the task required the agent to hold something between decisions, a belief about what is true, how certain it is, and what would change it, and the agent never formed it, never tested it, or revised it without acting on the revision.

The field has responded along three lines that rarely cite one another. Memory research moved the state from the reasoning trace (Wei et al., 2022b; Yao et al., 2023b) into external stores that persist, reflect, and abstract experience (Park et al., 2023; Shinn et al., 2023; Luo et al., 2026a). Uncertainty research taught models to state confidence (Lin et al., 2022; Kadavath et al., 2022) and extended it from single answers to agent trajectories (Lin et al., 2026b). Post-training moved

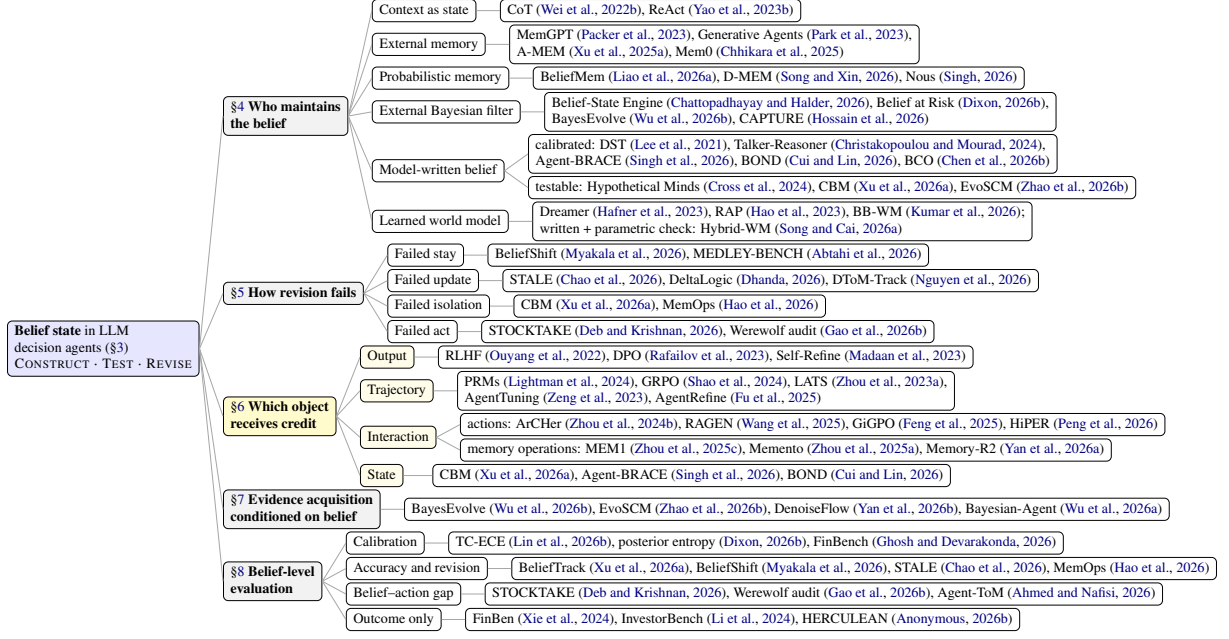


Figure 1: Taxonomy of the survey with representative systems by year. The credited object (yellow) is the axis every method is placed on; Table 12 in the appendix places every screened system.

from rewarding the finished response (Ouyang et al., 2022) to rewarding reasoning steps (Lightman et al., 2024; Zheng et al., 2025) and then actions over several turns of interaction (Zhou et al., 2024b; Zhang et al., 2025; Zhang, 2026). Each line supervises a different thing, what is stored, how confident the answer is, which action earns reward, and none of them supervises the belief itself.

Explicit beliefs are older than the recent attention suggests: dialogue state tracking wrote one in 2021 (Lee et al., 2021), agents wrote beliefs about tasks, opponents, and markets in 2024 (Kim et al., 2024; Cross et al., 2024; Yu et al., 2024), and two 2025 systems learned to overwrite a written state by reinforcement (Zhou et al., 2025c; Yu et al., 2025). Until recently the belief was credited, if at all, by the task outcome. In 2026 alone at least twelve systems gave it a learning signal or oracle of its own,¹ and benchmarks began to score the belief rather than the outcome it led to (Xu et al., 2026a; Deb and Krishnan, 2026; Chao et al., 2026), with disjoint formalisms and metrics and little mutual citation (timeline in Figure 7, appendix).

No existing survey takes this object as its own: memory surveys organise by what is stored, re-

ward and credit surveys stop at the action, uncertainty surveys score an output rather than a state, and evaluation surveys report no belief-level metric (Section 2, Table 1). This survey asks one question of every method: *which object receives credit*. Figure 1 gives the resulting taxonomy, and Figure 2 places the four answers on the observation–belief–action loop: the finished output, the sampled trajectory, the interaction (actions, tool calls, memory operations), or the belief written before the action and tested by evidence returned after it. It cuts across the neighbouring taxonomies rather than replacing them. We cover language-model agents acting over several steps under partial observability, from 2020 to September 2026, collected by a scripted protocol (Appendix A). Section 3 grounds the belief state in POMDP belief and belief-revision theory; Sections 4–8 organise the literature by representation, revision, credited object, evidence acquisition, and evaluation; Section 9 compares the four levels, and Section 10 states the open problems, chief among them that no published method obtains a signed signal for a written belief from the agent’s own predictions. Finance is the running example because delayed, partial evidence is its normal condition; the taxonomy is stated without reference to any domain.

¹Belief-State Engine, Belief at Risk and its validation framework, BayesEvolve, CAPTURE, Agent-BRACE, the Bayesian linguistic forecaster, BOND, CBM, BCO, EvoSCM, and the belief-based world model; Section 4.

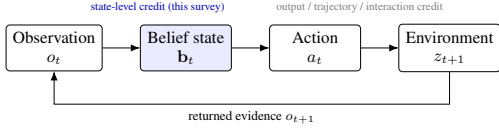


Figure 2: The observation-belief-action loop. Existing post-training credits the action, the trajectory that produced it, or the response text. State-level credit reaches the belief written before the action, using evidence returned after it.

2 Related Work

Table 1 places this survey among thirteen neighbouring ones, none of which takes the belief state as its object. Memory surveys (Zhang et al., 2024b; Hu et al., 2025b; Luo et al., 2026a) organise by what is stored and never ask whether a stored item is believed or what evidence would retire it. Reward and credit surveys (Wu, 2025; Zheng et al., 2025; Zhang et al., 2025; Zhang, 2026) follow the signal down to tokens, steps, and memory operations and stop at the action. Uncertainty surveys (Xia et al., 2025; Lin et al., 2026b) attach a score to an output rather than to a state carried forward. Evaluation, long-horizon, forecasting, and finance surveys (Yehudai et al., 2026; Chen et al., 2026a; Xu et al., 2026c; Dong et al., 2025b) treat belief as one component among many and report no metric that scores it apart from the outcome.

3 Problem Formulation

Partially observed decision-making defines the belief as the distribution over hidden variables that is sufficient for acting: with hidden state z_t , observation o_t , and history H_t , the belief is $\mathbf{b}_t = p(z_t | H_t)$, maintained by a Bayes filter (Åström, 1965; Kaelbling et al., 1998; Thrun et al., 2005). Agentic reinforcement learning casts the language-model agent in the same terms, a POMDP whose observable history is the context, with any explicit state kept in a working memory the model itself writes (Zhang et al., 2025; Sumers et al., 2023). In a ReAct-style agent each turn emits a reasoning segment and then an action (Yao et al., 2023b); the reasoning segment is where the state lives, and because the context only grows, the state is carried forward by being rewritten or re-read rather than by an estimator. We write $\mathbf{b}_t$ for whatever explicit state the span carries before the action a_t , and note that a later observation o_{t+1} can evaluate two different things: the action, and the belief that made

Who maintains the belief	Written	Uncert.	Testable	Auditable	Example
Context as state	●	○	○	○	ReAct
External memory	○	○	○	●	MemGPT
Probabilistic memory	○	●	○	○	Nous
External Bayesian filter	○	●	●	●	Belief-State Engine
Written, calibrated	●	●	○	○	Agent-BRACE, BOND
Written, testable	●	○	●	○	CBM, EvoSCM
Learned world model	○	○	●	○	Dreamer, BB-WM
Written + parametric check	●	●	●	○	Hybrid-WM

Table 2: Representation classes and the properties they trade off. *Written*: the model writes and reads the belief; *Uncert.*: it carries uncertainty; *Testable*: it commits to a falsifiable prediction; *Auditable*: it can be validated against a closed-form oracle. ● yes, ● partly, ○ no.

the action plausible.

Two older literatures say what such a belief must do. Bayesian filtering requires it to predict and to update on the mismatch (Thrun et al., 2005; Friston, 2010); belief-revision theory requires that a rational agent can expand its beliefs with a new claim, revise them when a claim is contradicted, and contract them without discarding the rest (Alchourrón et al., 1985). Read together, they give the three operations by which we place every system: CONSTRUCT, writing a claim and binding observations to it; TEST, committing to a prediction that a returned observation can falsify; and REVISE, changing the active belief without erasing what was believed. A *memory* stores traces and is judged by recall (Luo et al., 2026a); a *world model* predicts the next observation and is judged by prediction error (Li et al., 2026c); a *belief* interprets traces as claims about what is true now, with uncertainty, and is judged by calibration and by how fast it is revised when wrong. A reasonable action under a stale belief, the failure that motivates this survey, is invisible to both neighbours.

4 Belief Representation: Who Maintains the State

Where does the belief live, who writes it, and can it be wrong in a way that later evidence can show? We order representations by *who maintains the belief*, from no one to an explicit estimator, and by whether the belief is *testable*, that is, commits to a prediction that a returned observation can falsify. Table 2 summarises the classes; Figure 6 and Table 12 in the appendix place every system on both axes.

4.1 Context and Memory

The most common design has no representation at all: the reasoning trace is the state (Wei et al., 2022b; Yao et al., 2023b), the belief implicit and its uncertainty unstated. In a text game, agents that read the simulator’s latent state win about 79% of episodes against 11% for agents that read only observations (Alaswad et al., 2026), and on STALE the best model resolves old-versus-new conflicts in 55% of cases (Chao et al., 2026), and programmatic state abstraction alone helps substantially (Bogdanov et al., 2026; He et al., 2026a).

Memory-augmented agents moved the state into a store retrieved into context (Packer et al., 2023; Park et al., 2023; Zhong et al., 2024; Xu et al., 2025a), the evolution that Luo et al. (2026a) survey. Two limits matter for belief: the stored unit is a trace rather than a claim with a probability, and an update is an operation on the store rather than on a hypothesis, so answer accuracy hides failures in the update and forget operations themselves (Hao et al., 2026). Structured stores narrow the first limit with schemas, graphs, or bitemporal transitions (Xu and Fekri, 2026; Luo et al., 2026b; Liu et al., 2026b; Xu, 2026; Ning et al., 2026); probabilistic stores narrow the second by attaching a number to each claim (Liao et al., 2026a; Song and Xin, 2026; Yang et al., 2026b). Even closed-form Bayesian updates per stored claim barely beat last-write-wins on LoCoMo (Singh, 2026; Srivastava, 2026).

4.2 External Bayesian Filters

A different line takes the belief out of the model. The model maps evidence to a likelihood over a fixed set of latent regimes, hypotheses, or user states, and an explicit filter maintains the posterior, whose entropy and drift can then be reported as model risk (Chattopadhyay and Halder, 2026; Dixon, 2026b,c,a; Wu et al., 2026b; Hossain et al., 2026; Soru, 2026). These are the only beliefs in the survey validated against a closed-form oracle, and they are auditable by construction. The price is a hypothesis space fixed in advance: the filter cannot introduce a factor it was not given.

4.3 Model-Written Beliefs

The third line keeps the belief in context but gives it structure the model writes and reads. Dialogue state tracking wrote a slot-value belief in 2021 (Lee et al., 2021), verbalised confidence attached

a probability to an answer (Lin et al., 2022; Kada-vath et al., 2022), 2024 agents wrote beliefs about tasks, opponents, and markets (Kim et al., 2024; Christakopoulou and Mourad, 2024; Cross et al., 2024; Yu et al., 2024), and in 2025 MEM1 and MemAgent learned to overwrite a fixed written state each turn (Zhou et al., 2025c; Yu et al., 2025). The newest systems add uncertainty and verification to the written object, from certainty tags and Bayesian-teacher posteriors to symbolic verification and populations of causal models (Singh et al., 2026; Murphy, 2026; Cui and Lin, 2026; Xu et al., 2026a; Chen et al., 2026b; Seo et al., 2026; Giovannelli and Kent, 2026; Zhao et al., 2026b).

Testability separates them, and we define it operationally: a written belief is testable if it contains a statement about a future observation with a horizon, so that the returned observation yields a signed error for that statement. By this definition most written beliefs are calibrated but not testable: they state confidence or revise the belief but never name the observation that would falsify it. Hypothetical Minds, CBM, and EvoSCM do, and Section 6 turns on this distinction.

4.4 Learned World Models

The fourth line learns the dynamics and exposes the resulting belief to the policy: recurrent latent states (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2023), the language model as its own world model for tree search (Hao et al., 2023; Zhao et al., 2023b; Zhou et al., 2023a), LLM-proposed POMDPs refined to a belief-based likelihood (Six et al., 2026), and a world-model belief the policy can query (Kumar et al., 2026; Li et al., 2026c; Bagaria et al., 2026). World-model beliefs are testable by construction but, unless written back, cannot be reasoned about in language. Two 2026 systems close that gap by using the learned model as a check on a written belief rather than a replacement: a parametric consistency gate that cuts the hallucinated-state rate from 0.176 to 0.035 (Song and Cai, 2026a), and semantic rules distilled from prediction–observation mismatches (Zhang et al., 2026c).

Insight 1. Structure is not testability. Memory design has made the belief steadily more structured without making it falsifiable, and the representations that can be falsified, filters and world models, cannot be reasoned about in language. The belief the field needs is written and testable at once; no peer-reviewed system has one yet.

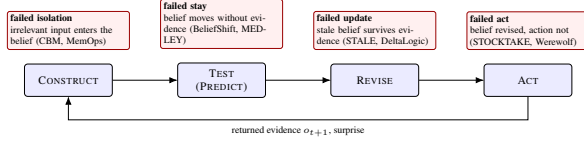


Figure 3: Where the four failure modes of Section 5 occur on the construct–test–revise–act cycle. Only a testable belief lets returned evidence be routed to the operation that failed rather than to the action.

Failure mode	Definition	Measured by	Headline result
Failed stay	belief moves without evidence	BeliefShift, MEDLEY-BENCH	consensus sensitivity from near 0 to large across 35 models
Failed update	belief survives contradicting evidence	DeltaLogic, DToM-Track, STALE	66.7% on a conclusion, 46.7% on whether an edit changes it
Failed isolation	irrelevant input enters the belief	BeliefTrack, MemOps	belief-state reward cuts failures by 71%
Failed act	belief revised, action unchanged	STOCKTAKE, Werewolf audit	34–43% of correctly diagnosed weeks end in the wrong action

Table 3: Four failure modes of belief revision, the benchmarks that isolate each, and their headline results (Myakala et al., 2026; Abtahi et al., 2026; Dhanda, 2026; Nguyen et al., 2026; Chao et al., 2026; Xu et al., 2026a; Hao et al., 2026; Deb and Krishnan, 2026; Gao et al., 2026b).

5 Belief Revision and Its Failure Modes

Bayesian filtering and active inference agree on how a belief should change: it predicts, evidence arrives, and the mismatch drives the update (Friston, 2010; Friston et al., 2017); factor graphs then assign the error to the factors that produced the prediction (Kschischang et al., 2001; Loeliger, 2004). The language-model literature imported the vocabulary but rarely the routing: feedback is appended to the context or folded into self-critique loops that revise an output, a plan, or a trajectory (Huang et al., 2022b; Madaan et al., 2023; Chen et al., 2023b; Kim et al., 2023; Wang et al., 2023b).

5.1 Four Failure Modes

Contextual Belief Management names three ways a belief can fail on new input, a *failed stay* in which the belief moves without evidence, a *failed update* in which it survives evidence against it, and a *failed isolation* in which irrelevant input contaminates it (Xu et al., 2026a). Three independent studies document a fourth, the *failed act*, in which the belief is revised but the action is not, so that a belief-level oracle finds the diagnosis right while

the action stays wrong (Deb and Krishnan, 2026; Gao et al., 2026b). Table 3 lists the benchmark that isolates each mode; their placement on the construct–test–revise–act cycle (Figure 3) is ours, and each names the operation whose signal should receive the error.

5.2 Mechanisms of Revision

Failed update is now a benchmark topic in its own right (Chao et al., 2026; Sun and He, 2026; Nakayashiki, 2026; Yang et al., 2026a); in finance it is a stale factor, visible as drift in the agent’s planning embeddings before the trading loss appears (Xue, 2026). The mechanisms that respond fall into three families by what triggers them. The first is triggered by *contradiction* among stored claims and resolves it with an operator algebra, first-class update and forget operations, or super-session contracts that keep the superseded claim visible (Wang, 2026; Hao et al., 2026; Xu, 2026). The second is triggered *before commitment*, by self-audit, a parametric veto, semantic entropy, or resampling (Yuan et al., 2026; Song and Cai, 2026a; Yan et al., 2026b; Wang et al., 2026b); these see no future observation and so cannot catch a belief that is consistent and wrong. The third is triggered by *surprise*: reward-prediction error routes writes to a fast or a slow path (Song and Xin, 2026), and prediction–observation mismatches become semantic rules (Zhang et al., 2026c).

Routing the error to the claim that produced the prediction, the factor-graph step with which this section opened, is what none of the surveyed systems implements in the model’s own context.

Insight 2. Revision is triggered, not routed. Contradiction, self-audit, and surprise decide *when* a belief changes; none decides *which claim* the error belongs to. Until the error is routed to the claim that made the prediction, agents will keep repairing memories while beliefs stay wrong, and failed act will remain the unmeasured mode.

6 Learning Signals by Credited Object

Existing surveys organise post-training by where the reward comes from, where the trajectories come from, and how dense the signal is (Wu, 2025; Zhang, 2026; Zheng et al., 2025). We hold those fixed and vary one thing, the object that receives the gradient (Figure 4, Table 4). The criterion is operational: a method is at the *state level* if the quantity that receives a gradient is a claim about the environment and its sign depends on an observation returned after the action. A method that rewards a memory write regardless of whether the

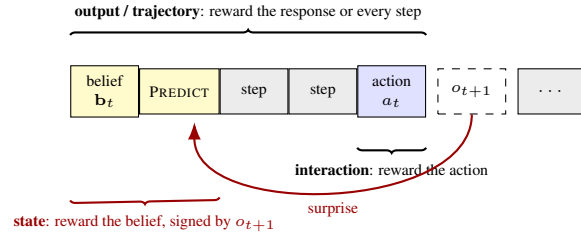


Figure 4: The credited object on one span of a trajectory. Output- and trajectory-level credit reaches text generated from the context; interaction-level credit reaches the action; state-level credit reaches the written belief, and its sign comes from the observation returned after the action.

Level	Credited object	Signed by o_{t+1} ?	Representative
Output	finished response	no	RLHF, DPO
Trajectory	sampled steps	no	PRMs, GRPO, LATS
Interaction	actions, tool calls	no (task reward)	ArCher, RAGEN, HiPER
Interaction	memory operations	no (operation reward)	MEM1, Memento, Memory-R2
State	written belief	closed-world oracle	CBM
State	claim + certainty	calibration target	Agent-BRACE
State	posterior tag	Bayesian teacher	BOND
State	written belief	<i>own prediction</i>	(none published)

Table 4: Learning signals by credited object and where the sign of the signal comes from (Section 6); the full version with feedback source, granularity, and evidence level is Table 10 in the appendix.

written claim was true is at the interaction level, however dense its credit.

6.1 Output and Trajectory

Supervised fine-tuning and preference optimisation credit the completed response (Ouyang et al., 2022; Wei et al., 2022a; Rafailov et al., 2023), and self-consistency and self-improvement select the responses to train on (Wang et al., 2022; Huang et al., 2022a). Trajectory-level credit began by imitating and searching over agent trajectories (Chen et al., 2023a; Zeng et al., 2023; Xi et al., 2024; Zhou et al., 2023a; Putta et al., 2024; Fu et al., 2025), then credited sampled steps directly through PPO, GRPO, and process reward models (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024; Zheng et al., 2025; Zhang et al., 2026e; Tan et al., 2026).

6.2 Interaction

Agentic reinforcement learning credits actions over several turns (Zhou et al., 2024b; Qi et al., 2024; Wang et al., 2025; Chen et al., 2025a; Zhou et al., 2025b; Feng et al., 2025; Dong et al., 2025a), and the open question is how to make that credit denser without making it wrong, by hierarchical advantages, counterfactual estimators over

Acquisition conditioned on	conditioned on	Systems	Belief object
fixed heuristic		agentic RAG (Gao et al., 2024)	none
task-level POMDP		CGDP (Kausik et al., 2026), MedExAgent (Gao et al., 2026a)	searcher, patient state
uncertainty of an explicit belief		BayesEvo (Wu et al., 2026b), EvoSCM (Zhao et al., 2026b), DenoiseFlow (Yan et al., 2026b)	hypothesis posterior, causal population, step entropy
belief over own resources		Bayesian-Agent (Wu et al., 2026a), REDEREF (Hosseini et al., 2026), CMI (Srivastava, 2026)	skill, delegate, memory reliability
information gain as reward		PiCA (Liu et al., 2026a), TIPS (Xie et al., 2026)	implicit
written testable belief		<i>none found</i>	—

Table 5: Evidence-acquisition decisions by what they are conditioned on.

shared states, potential-based shaping, or verifier-structured credit (Peng et al., 2026; Wang et al., 2026a; Diao et al., 2026; Wang et al., 2026f; Ren et al., 2026; Liao et al., 2026b; He et al., 2026b; Liu et al., 2026a; Xie et al., 2026; Zheng et al., 2026; Zhang et al., 2026b; Li et al., 2026a; Wang et al., 2026g; Guo et al., 2026; Qi et al., 2026; Zhao et al., 2026a; Lin et al., 2026a); denser is not free (Modecrua et al., 2026; Pu et al., 2026). The nearest this level comes to the state is a group that credits the write to a memory, from a rewritten case bank to an RL-trained overwrite of a fixed written state (Zhou et al., 2025a,c; Yu et al., 2025; Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026; Wang and Jiang, 2026; Hu et al., 2026). By the criterion above these remain interaction-level: they credit the *operation*, so a well-executed update to a wrong claim is still rewarded.

6.3 State

Three published methods meet the criterion. CBM trains with belief-state rewards from symbolic verification and cuts belief-management failures by about 71% (Xu et al., 2026a). Agent-BRACE trains belief model and policy jointly with verbalised certainty as the belief’s output (Singh et al., 2026). BOND distils a Bayesian teacher’s posterior into an 8B student whose Brier score beats a 70B prompted baseline (Cui and Lin, 2026).

Insight 3. Credit has been getting denser, not deeper. Agentic reinforcement learning has refined which action to prefer; no method changed what the action is conditioned on. The two axes are orthogonal, and the first system to combine them will need a sign for its belief that the environment, not an oracle, supplies.

	calibration	accuracy	revision	belief-action	outcome
UQ taxonomy (Lin et al., 2026b) A	•	◦	◦	◦	◦
Agent-BRACE (Singh et al., 2026) A	•	◦	◦	◦	◦
Belief-State Engine (Chattopadhyay and Halder, 2026) A	•	◦	◦	•	◦
Belief at Risk (Dixon, 2026b) A	•	◦	◦	◦	◦
FinBench (Ghosh and Devarakonda, 2026) A	•	◦	◦	◦	•
BeliefTrack (Xu et al., 2026a) A	◦	•	•	◦	◦
DeltaLogic (Dhanda, 2026) A	◦	•	•	◦	◦
DToM-Track (Nguyen et al., 2026) A	◦	•	•	◦	◦
BeliefShift (Myakala et al., 2026) A	◦	◦	•	◦	◦
STALE (Chao et al., 2026) A	◦	◦	•	•	◦
MemOps (Hao et al., 2026) A	◦	◦	•	◦	◦
STOCKTAKE (Deb and Krishnan, 2026) A	◦	•	◦	•	•
Werewolf audit (Gao et al., 2026b) A	◦	◦	◦	•	•
HERCULEAN (Anonymous, 2026b) A	◦	◦	◦	◦	•
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024) P	◦	◦	◦	◦	•

Figure 5: Which belief-level construct each evaluation paper reports. Filled dots mark a reported metric. P = peer-reviewed venue, A = arXiv preprint at the time of writing. No two papers report the same metric for the same construct, which is the fragmentation this section addresses.

7 Belief-Guided Evidence Acquisition

A testable belief tells the agent what evidence would change it, which turns retrieval, tool use, and routing from fixed pipeline stages into decisions (Table 5). Agentic search (Kausik et al., 2026) and clinical diagnosis (Gao et al., 2026a) have been framed as POMDPs whose actions include what to observe next, and two finance simulators show why it matters: failing agents spend 62% of their observation budget re-reading internal state, and failures trace to incomplete acquisition rather than to reasoning (Han et al., 2026; Zhang et al., 2026a).

The methods that condition acquisition on a belief differ in what the belief is over (Table 5). Agentic retrieval first let the model decide when to retrieve (Gao et al., 2024); recent systems condition that decision on measured uncertainty about the task (Wu et al., 2026b; Zhao et al., 2026b; Xu and Fekri, 2026; Yan et al., 2026b), make information gain the training signal itself (Liu et al., 2026a; Xie et al., 2026), or keep a belief over the agent’s own skills, delegates, or memories (Wu et al., 2026a; Hosseini et al., 2026; Srivastava, 2026). Financial evidence adds a constraint, since much of it cannot leave its silo; knowledge harnesses and federated adaptation are the compliance-aware forms of integration (Borjigin et al., 2026; Anonymous, 2026a).

Insight 4. Acquisition is a belief problem. Agents lose by re-reading what they know rather than seeking what would change it. A testable belief names that evidence, so evidence acquisition will move from a fixed retrieval stage to a consequence of the belief.

8 Evaluating Belief Apart from Outcome

A realised trade or a correct answer can follow from a correct belief, from a wrong belief that did not matter, or from a belief revised too late, and outcome metrics cannot separate the three; this is why stronger backbones do not yield stronger trading agents and the benchmarks cannot say why (Xie et al., 2024; Li et al., 2024). Earlier agent benchmarks score task success or recall (Shridhar et al., 2020; Liu et al., 2023; Maharana et al., 2024; Yao et al., 2024); recent work responded with belief-level metrics, each paper defining its own. Figure 5 aligns them by construct, and Table 11 in the appendix catalogues 33 benchmarks by whether the evaluator can observe the latent state.

8.1 Constructs

Calibration over a trajectory has the most statistics and the least comparability: checkpoint calibration error, episode-level calibration, posterior entropy, and proper scoring rules on time-gated forecasts express one construct in four ways, none computed on another paper’s system (Lin et al., 2026b; Singh et al., 2026; Chattopadhyay and Halder, 2026; Dixon, 2026b; Ghosh and Devarakonda, 2026). *Belief accuracy and revision* are measured against a known answer in a closed world, after a minimal premise edit, and across recall, inference, and change detection (Xu et al., 2026a; Dhanda, 2026; Nguyen et al., 2026), and against human annotation (Myakala et al., 2026; Chao et al., 2026; Hao et al., 2026; Yang et al., 2026a) and, for beliefs about other agents, by theory-of-mind benchmarks (Bawatneh et al., 2026; Wang et al., 2026c). The *belief-action gap* is the youngest construct: STOCKTAKE scores the agent against an exact Bayes filter on the same observations (Deb and Krishnan, 2026), the Werewolf audit reports action-belief consistency (Gao et al., 2026b), and Agent-ToM monitors an agent’s beliefs from its trajectory (Ahmed and Nafisi, 2026).

8.2 Where the Latent State Comes From

Every construct beyond calibration needs a latent state to compare against. Controlled environments provide it (Deb and Krishnan, 2026; Xu et al., 2026a; Alaswad et al., 2026); long-horizon simulators know it but report belief only through proxies (He et al., 2026a; Zhang et al., 2026a; Han et al.,

Level	Sign of the signal from	Env.	No oracle	Belief	Act gap	Maturity
Output	humans, verifiers	○	●	○	○	●
Trajectory	verifier, PRM	●	●	○	○	●
Interaction (actions)	task reward	●	●	○	○	●
Interaction (memory ops)	task reward, teacher	●	●	○	○	○
State (oracle or teacher)	closed world, teacher, target	○	○	●	●	○
State (own prediction)	returned observation	●	●	●	●	○

Table 6: The four credit levels compared. *Env.*: the sign of the signal comes from the environment; *No oracle*: no external source of truth is needed; *Belief*: the written belief receives the gradient; *Act gap*: belief error and action error are separated; *Maturity*: ● many systems, ● a few, ○ none published.

2026); historical replay provides neither, so workflow benchmarks can report a belief failure only at the outcome level (Anonymous, 2026b).

Insight 5. Belief cannot be scored by outcome. Five constructs, fifteen metrics, no cross-paper comparison; only the fair oracle scores belief and action on the same episodes. Progress on belief will be measurable only where the latent state is known, so controlled environments, not replay, are where the next results should come from.

9 Discussion

Table 6 compares the four credit levels along the properties the preceding sections exposed; three observations follow.

The sign is the scarce resource. Output and trajectory levels take the sign of their signal from humans or verifiers, the interaction level from the task reward, and the state level, in all three published cases, from an oracle or teacher the environment does not supply. Denser credit multiplies a sign the environment already gives; state credit needs a sign for a claim, and without an oracle that sign can only come from the agent’s own prediction settled against the next observation. That row is empty.

Auditability and expressiveness pull apart. Filters are validated in closed form but cannot introduce a factor they were not given; written beliefs can name a new factor but are rarely testable; world models are testable but mute. The field is converging on hybrids that write the belief and check it parametrically (Song and Cai, 2026a; Zhang et al., 2026c; Kumar et al., 2026).

The belief–action gap is a training problem. Wherever belief and action are scored on the same episodes, prompting and oracles improve the belief while action–belief consistency barely moves (Deb and Krishnan, 2026; Gao et al., 2026b); the one system that moves both trains belief and policy jointly (Singh et al., 2026). A policy conditioned

on a stale belief does not change when the belief is corrected in context, so closing the gap requires the state-level and interaction-level signals of Table 6 in one objective, not a better belief alone.

10 Open Problems

Self-supervised testability. The state-level methods need an external source of truth, and the surprise-triggered mechanisms of Section 5 have the ingredient that would remove that need but attach the error to a memory entry rather than to the written claim. Combining the two, so that the agent’s own matured predictions supply the signed signal for its written belief, is the most direct open problem this survey identifies. Which evidence horizon to commit to (Ghosh and Devarakonda, 2026; Zhao et al., 2026b) and how much superseded belief must stay visible (Singh et al., 2026; Diao et al., 2026) are its open parameters.

Latent beliefs without ground truth. Every construct beyond calibration needs an oracle, a closed world, or annotation; historical data has none. Filter-recovered latents and representation drift are candidate substitutes, unvalidated against each other (Dixon, 2026b; Xue, 2026); a benchmark pairing a controlled environment with a replay of the same instrument would settle whether belief-level gains transfer.

Belief without action, and belief under governance. Revising the belief does not revise a policy conditioned on the old one, and action–belief consistency sits far below belief accuracy wherever both are measured (Gao et al., 2026b; Deb and Krishnan, 2026). Regulated domains add that the belief must be auditable, which today only fixed-space filters provide; LLM-proposed spaces (Six et al., 2026; Soru, 2026; Chen et al., 2026b) are the candidate bridge (Dixon, 2026c).

11 Conclusion

This survey asked one question of the literature on language-model decision agents: which object receives credit. Organising 179 papers by the answer exposes a field that has moved the belief from the trace into stores, filters, and world models, learned to make credit denser at every level below the state, and built fifteen ways to measure belief without a shared one. Three findings carry across the axes. The representations that can be falsified are not the

ones the agent can reason about; the mechanisms that revise beliefs do not route the error to the claim that produced it; and the three methods that credit the belief at all borrow the sign of their signal from an oracle the environment does not supply. The gap they leave is precise, and we argue it is the next step: an agent whose own matured predictions supply the signed signal for its written belief, evaluated where the latent state is known. The taxonomy, screening record, and benchmark catalogue are released to make that step measurable.

Limitations

This survey covers work indexed by OpenAlex and arXiv from January 2020 to 15 September 2026 and retrieved by the keyword protocol in Appendix A; papers that describe belief maintenance without using any of our search terms may be missed. Our taxonomy assigns each method to one representation class and one credit level, although several systems combine classes. The survey reports numbers as stated by the cited papers and does not re-run any method; headline figures from preprints may change in later versions. Finance is used as the running example because delayed and partially observed evidence is the norm there; the taxonomy itself is domain-agnostic, but we have not validated it against embodied or dialogue agents in the same depth.

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A Paper Collection Protocol

The search is scripted and rerunnable; the configuration, raw hits, and per-stage counts for the runs reported here (identifiers 20260915_v1 for 2025–2026 and 20260916_v3 for the 2020–2026 window) are released with the paper.

Queries. Twenty-three keyword queries were defined in the first run, four to five per survey section; a second run added eleven queries on evidence acquisition and theory-of-mind belief modelling (34 in total): representation (Section 4), revision (Section 5), learning (Section 6), evidence (Section 7), and evaluation (Section 8).

Sources. Source A queried OpenAlex across all venues with publication date from 1 January 2025, 50 results per query. Source B queried OpenAlex restricted to the arXiv source with publication date from 1 January 2026, 50 results per query, to improve recall of preprints. The arXiv API itself was rate-limited during the run; the OpenAlex mirror of arXiv lags by a few days.

Screening. Records were deduplicated by identifier and then by normalised title, so that a preprint and its journal version count once. Rule-based screening on title and abstract applied one inclusion criterion for an LLM-related term (I1), one exclusion criterion for off-domain applications (E1), and one inclusion criterion requiring at least two topic terms from a fixed list (I2). Records without an abstract of at least 200 characters were excluded. Included records were ranked by a relevance score used only for ordering the manual full-text screen, never for inclusion.

Second and third runs. The second run (34 queries, same sources and rules) returned 834 unique records, 338 after screening, of which 96 were published in 2026; only two of those had not already been read in the first run, which we take as evidence that the first run had saturated the keyword neighbourhood. The third run extended the window to 1 January 2020 with 100 results per query and returned 4,471 unique records, 3,999 after title deduplication and 2,079 after rule screening. Keyword ranking is dominated by generic high-citation papers before 2025, so the 2020–2025 foundations were enumerated as a curated

Stage	Count
Queries	23
Source A raw hits / unique	1,150 / 715
Source B raw hits / unique	1,150 / 1,008
Merged unique (identifier)	1,685
After title-level deduplication	1,460
Excluded: no or short abstract	45
Excluded I1: no LLM term	160
Excluded E1: off-domain	35
Excluded I2: fewer than two topic terms	357
Included after rule screening	863
of which published in 2026	656
Full-text screened	117
of which included after reading	106

Table 7: Paper-collection counts for run 20260915_v1.

list of 90 titles resolved through OpenAlex for venue and citation data (Table 13); 75 of them were read in full and 73 included.

Evidence level. Each cited source is labelled P (peer-reviewed venue) or A (arXiv preprint at the time of writing) in the tables and in Figure 5. Of the 179 papers included after full-text reading, 47 are peer-reviewed and 132 are preprints; every number quoted in the text is the paper’s own headline figure, taken from the abstract or the reported table, and none has been reproduced by us.

Full-text screening. Each paper read in full was recorded with the fields: who maintains the belief, credited object, revision signal, whether belief-level metrics are reported, survey section, and role. The record is the source of Tables 8–11 and Figure 5.

B Additional Tables and Figures

C Timeline of Credited Objects

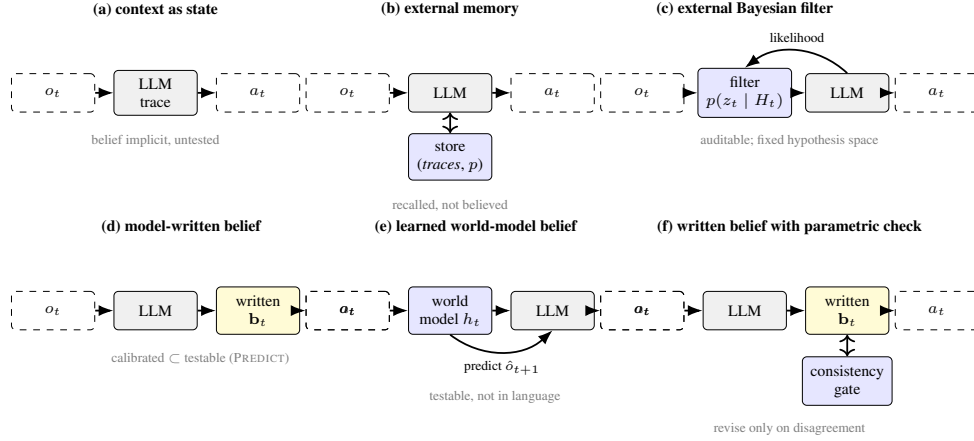


Figure 6: Who maintains the belief. Yellow marks a belief the model writes and can reason about; blue marks an external estimator. Only (c), (d)-testable, (e), and (f) commit to a prediction that later evidence can falsify.

Class	Method	Model-written	Append-only	Uncertainty	TEST	REVISE	Domain
Context as state	ReAct (Yao et al., 2023b)	✓	✓	✗	✗	✗	general
External memory	MemGPT (Packer et al., 2023)	✗	✗	✗	✗	store update	dialogue
Probabilistic memory	BeliefMem (Liao et al., 2026a)	✗	✗	prob.	✗	noisy-OR	LoCoMo, ALFWorld
Probabilistic memory	D-MEM (Song and Xin, 2026)	✗	✗	RPE	✗	gated write	LoCoMo
External filter	Belief-State Engine (Chattopadhyay and Halder, 2026)	✗	✗	posterior	✓	Bayes	Tiger, security
External filter	Belief at Risk (Dixon, 2026b)	✗	✗	posterior	✓	Bayes	equities
Model-written	Agent-BRACE (Singh et al., 2026)	✓	✗	ordinal	✗	RL	ALFWorld-style
Model-written	Linguistic forecaster (Murphy, 2026)	✓	✓	numeric	✓	sequential	ForecastBench
Model-written	BOND (Cui and Lin, 2026)	✓	✓	posterior tag	✗	distilled	negotiation
Model-written	CBM (Xu et al., 2026a)	✓	✓	✗	✓(symbolic)	RL	closed world
Model-written	EvoSCM (Zhao et al., 2026b)	✓	✓	population	✓(interventions)	rule distillation	science
World model	BB-WM (Kumar et al., 2026)	✗	✗	latent	✓	learned	general

Table 8: Belief representation by who maintains the belief. TEST: the belief commits to a falsifiable prediction. All 2026 entries are arXiv preprints at the time of writing; ReAct and MemGPT are peer-reviewed.

Method	Failed stay	Failed update	Failed isolation	Failed act	Trigger	Signal projected to
Reflexion (Shinn et al., 2023)		✓			task failure	whole context
BeliefShift (Myakala et al., 2026)	✓	✓			user evidence	(benchmark)
STALE (Chao et al., 2026)		✓		✓	implicit conflict	(benchmark)
CBM (Xu et al., 2026a)	✓	✓	✓		symbolic check	belief state
TOKI (Wang, 2026)		✓			contradiction	memory rows
MemOps (Hao et al., 2026)		✓	✓		lifecycle op	(benchmark)
D-MEM (Song and Xin, 2026)		✓	✓		reward pred. error	memory path
STOCKTAKE (Deb and Krishnan, 2026)		✓		✓	oracle mismatch	(benchmark)
Agent-BRACE (Singh et al., 2026)		✓		✓	certainty	belief model
Werewolf audit (Gao et al., 2026b)		✓		✓	external belief log	(benchmark)

Table 9: Belief revision methods and benchmarks by failure mode addressed, revision trigger, and where the signal is projected.

Level	Representative methods	Feedback source	Granularity	Signed by returned o_{t+1}	Delayed evidence	Partial obs.	Evidence
Output	SFT, RLHF, DPO (Ouyang et al., 2022; Rafailov et al., 2023)	demos, preferences	response	✗	✗	✗	P
Trajectory	PPO, GRPO, PRMs (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024)	verifier, PRM	step / rollout	✗	✗	✗	P
Interaction	HiPER, G ² PO, ECPO, IAPPO (Peng et al., 2026; Wang et al., 2026f; Li et al., 2026d; Ren et al., 2026)	environment	turn / action	✗	partial	✗	A
Interaction	Memory-R2, TAMTRL, EvoHarness-RL (Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026)	environment, teacher	memory operation	✗	partial	✓	A
State	CBM (Xu et al., 2026a)	symbolic belief reward	belief state	closed world	✓	✓	A
State	Agent-BRACE (Singh et al., 2026)	joint RL, calibration	claim + certainty	calibration target	✓	✓	A
State	BOND (Cui and Lin, 2026)	Bayesian teacher	posterior tag	teacher	✗	✓	A

Table 10: Learning signals by credited object. Evidence: P = peer-reviewed venue, A = arXiv preprint at the time of writing.

Benchmark	Domain	Horizon	Latent state	Observable to evaluator	Delayed evidence	Belief-level quantity reported
ALFWorld (Shridhar et al., 2020)	text + embodied	episodic	household state	✓	partial	none (task success)
PlanBench (Valmeekam et al., 2022)	classical planning	episodic plan-length	PDDL state	✓	✗	plan verification, replanning
AgentBench (Liu et al., 2023)	8 environments	episodic	environment state	✓	partial	none (task success)
SmartPlay (Wu et al., 2023)	6 games	episodic	game state	✓	partial	none (9 capability scores)
T4D / FaR (Zhou et al., 2023b)	theory of mind	single-turn	others' beliefs	✓(annotated)	✗	belief tracking vs. action selection
False-belief tasks (Kosinski, 2023)	theory of mind	single-turn	others' beliefs	✓(annotated)	✗	false-belief accuracy
LoCoMo (Maharana et al., 2024)	long conversations	35 sessions	conversation facts	annotated	✓	temporal QA (proxy)
7-bench / r^2 -bench (Yao et al., 2024; Barres et al., 2025)	tool-user interaction	multi-turn	database state	✓	✓	pass-% consistency; reasoning vs. communication errors
MemoryAgentBench (Hu et al., 2025a)	incremental multi-turn	long-horizon	memory contents	annotated	✓	retrieval, test-time learning, conflict resolution
BeliefTrack (Xu et al., 2026a)	closed-world reasoning	multi-turn	finite belief space	✓(symbolic)	✓	turn-level belief accuracy; failed state/update/isolation
DeltaLogic (Dhanda, 2026)	logical reasoning	2 steps	premise set	✓	✓	revision accuracy vs. initial accuracy
DTOM-Track (Nguyen et al., 2026)	dialogue	multi-turn	interlocutor belief	✓(annotated)	✓	prior-belief recall; change detection
BeliefShift (Myakala et al., 2026)	user modelling	multi-session	user opinion	annotated	✓	revision accuracy; drift coherence; evidence sensitivity
OmniTOM (Bawaneh et al., 2026)	theory of mind	narrative	actors' beliefs	✓(annotated)	✓	belief-proposition extraction; schema labels
MINDGAMES (Wang et al., 2026c)	social games	repeated games	hidden roles, opponents	✗	✗	TrueSkill; turn-level logs
MEDLEY-BENCH (Abbahi et al., 2026)	social pressure	2 rounds	—	✗	✗	sensitivity to manipulated consensus
STALE (Chao et al., 2026)	memory validity	multi-turn	fact validity	annotated	✓	state resolution; premise resistance; policy adaptation
MemOps (Hao et al., 2026)	long conversations	multi-session	memory lifecycle	annotated	✓	operation-level probes
GroupMemBench (Yang et al., 2026a)	multi-party dialogue	multi-session	per-speaker knowledge	annotated	✓	information-update accuracy
MemGym (Xu et al., 2026b)	mixed agent tasks	long-horizon	—	✗	partial	memory-isolated score delta
STOCKTAKE (Deb and Krishnan, 2026)	inventory control	weekly, long	demand factors	✓(Bayes filter)	✓	detection lag; skill vs. oracle; knowing-doing rate
Flux (Alawad et al., 2026)	text game	Reservoir	simulator state	✓	✓	state-tracking error modes
YC-Bench (He et al., 2026a)	startup simulation	hundreds of turns	market, clients	privileged outcome	✓	scrapped usage (proxy)
RetailBench (Zhang et al., 2026a)	retail POMDP	180 days	demand, supply	privileged oracle	✓	evidence-acquisition behaviour (proxy)
EnterpriseArena (Han et al., 2026)	lending	132 months	borrower pool	privileged outcome	✓	observation-budget allocation (proxy)
Business Arena (Pan et al., 2026)	marketplace	long-horizon	market	privileged outcome	✓	action-level attribution
Werewolf audit (Gao et al., 2026b)	social deduction	9-player game	roles	✓	✓	action-belief consistency
CAGE-2 study (Bogdanov et al., 2026)	cyber defence	episodic	network state	✓	partial	state-abstraction return gain
MDS state tracking (Pai and Xian, 2026)	tool chains	196 calls	registers	✓	✗	exact-state success
FinBench (time-gated) (Ghosh and Devarakonda, 2026)	asset forecasting	multi-week	returns	realised	✓	Brier; interval scores
TradingArena (Xue, 2026)	trading	multi-week	regimes	realised	✓	representation drift (early warning)
HERCULEAN (Anonymous, 2026b)	finance workflows	multi-step	market, books	partial	✓	state-tracking sub-score
FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024)	finance tasks / trading	task / episodic	—	✗	partial	none (outcome only)

Table 11: Benchmarks that exercise belief maintenance, by whether the evaluator can observe the latent state and what belief-level quantity is reported. “Annotated” means human labels stand in for the latent state; “privileged” means the simulator knows it but the benchmark reports only outcomes. Ev. column omitted for space; peer-reviewed entries are ALFWorld, PlanBench, AgentBench, SmartPlay, T4D, false-belief tasks, LoCoMo, FinBen, and InvestorBench; the rest are arXiv preprints at the time of writing.

System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.	System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.
Schema-driven DST (Lee et al., 2021)	2021	written	none	dialogue history plus schema...	✓	P	Cyclic subtask graphs (Charzedeine and Saab, 2026)	2026	written	none	state-analysis-and-routing...	—	A
Verbalized Uncertainty (Lin et al., 2022)	2022	written	output	self-evaluation P and P of...	✓	A	CHAL (Giovannelli and Kent, 2026)	2026	written	none	gradient-informed revision on...	o	A
DrumV3 (Hafner et al., 2023)	2023	world model	world-model loss	RSSM latent posterior updated...	o	P	Predictor (Six et al., 2026)	2026	world model	world-model loss	belief-based likelihood score...	o	A
RecurrentGPT (Zhou et al., 2023c)	2023	written	output	LLM rewrites its own...	o	A	Belief Engine stance... (Yang et al., 2026b)	2026	prob. memory	none	log-odds update with evidence...	✓	A
LLM-MCTS (Zhou et al., 2023b)	2023	world model	trajectory	LLM-sampled world state...	o	P	HiPF (Diao et al., 2026)	2026	written	trajectory	subgoal completion +...	o	A
RAP (Hao et al., 2023)	2023	world model	trajectory	LLM-as-world-model predicted...	o	P	ProPlay (Ma et al., 2026)	2026	world model	trajectory	post-episode graph refinement...	o	A
Reflect-RL (Zhou et al., 2024a)	2024	written	interaction	reverse reflection model...	o	P	POMDP validation... (Dixon, 2026c)	2026	ext. filter	none	Bayesian	✓	A
Hypothetical Minds (Cross et al., 2024)	2024	written	none	hypotheses about other...	o	A	Internalizing the Future (Zhang et al., 2026d)	2026	written	trajectory	foreight-conditioned RL...	o	A
Talker Reasoner (Christakopoulos and Mourad, 2024)	2024	written	none	Reasoner writes/updates...	o	A	Hybrid-WM (Song and Cai, 2026a)	2026	world model	world-model loss	consistency gate	✓	A
QUBE (Kim et al., 2024)	2024	written	none	question-answering over...	o	A	Governance-aware POMDP... (Dixon, 2026a)	2026	ext. filter	none	Bayesian update on evidence...	o	A
MEMI (Zhou et al., 2025c)	2025	written	outcome (written state)	internal state rewritten each...	o	A	BayoEvolve (Wu et al., 2026b)	2026	ext. filter	none	posterior refit after each...	✓	A
MemAgent (Yu et al., 2025)	2025	written	outcome (written state)	task-wise overwrite of...	o	A	WorldEvolver (Zhang et al., 2026c)	2026	world model	world-model loss	prediction-observation...	o	A
PCE (Seo et al., 2026)	2026	written	none	scenario likelihood scoring...	o	A	WM-SAR (Song and Cai, 2026b)	2026	world model	trajectory	spectral error amplification	o	A
RB-VLA (Bagaria et al., 2026)	2026	world model	world-model loss	self-supervised world-model...	o	A	BCO (Chen et al., 2026b)	2026	written	none	evaluation of new candidate...	✓	A
REDEREF (Hossaini et al., 2026)	2026	prob. memory	interaction	judge feedback updates...	o	A	EvosCM causal belief... (Zhao et al., 2026b)	2026	written	none	prediction-observation...	✓	A
TAMTRL (Wang et al., 2026d)	2026	written	trajectory	teacher likelihood of memory...	o	A	CAPTURE (Hossaini et al., 2026)	2026	world model	world-model loss	uncertainty-triggered...	✓	A
Self-Guide internal... (Wang et al., 2026e)	2026	written	trajectory	self-generated guidance as...	o	A	Semantic Bayesian World... (Soru, 2026)	2026	ext. filter	none	Bayesian conditioning on...	o	A
ReflectChain... (Lao, 2026)	2026	world model	trajectory	latent trajectory rehearsal...	o	A	Belief-State Engine (Chattopadhyay and Halder, 2026)	2026	ext. filter	none (prompting)	Bayesian posterior	✓	A

Table 12: Systems with an explicit belief mechanism, by year. Belief kept by: who maintains the belief (Section 4); Credited: the object that receives the learning signal under the criterion of Section 6, none if the belief is not trained; Metric: ✓ reports a belief-level quantity, o a proxy; Ev.: P peer-reviewed, A arXiv preprint.

System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.	System	Year	Belief kept by	Credited	Revision trigger	Metric	Ev.
Decision Transformer (Chen et al., 2021)	2021	context	interaction	return-to-go conditioning...		P	HCAPO hindsight critic (Tan et al., 2026)	2026	context	trajectory	post-hoc LLM critic refinement		A
Schema-driven DST (Lee et al., 2021)	2021	written	none	dialogue history plus schema...	✓	P	TIPS (Xie et al., 2026)	2026	context	interaction	teacher answer-likelihood gain	o	A
Chain-of-Thought (Wei et al., 2022b)	2022	context	output	none		P	DTOM-Track (Nguyen et al., 2026)	2026	context	none	none		A
Verbalized Uncertainty (Lin et al., 2022)	2022	written	output	fine-tuning on calibrated...	✓	P	D-MEM (Song and Xian, 2026)	2026	ext. memory	none	reward prediction error gating	o	A
LMs Know (Kadavath et al., 2022)	2022	written	output	self-evaluation P and P of...	✓	A	SLEA-RL (Wang and Jiang, 2026)	2026	ext. memory	trajectory	observation change triggers...	o	A
Inner Monologue (Huang et al., 2022b)	2022	context	interaction	success detection, scene...		P	Graph of States (Lao et al., 2026b)	2026	ext. memory	none	symbolic check	o	A
DrumV3 (Hafner et al., 2023)	2023	world model	world-model loss	RSSM latent posterior updated...	o	P	TAMTRL (Wang et al., 2026d)	2026	written	trajectory	teacher likelihood of memory...	o	A
DEPS (Wang et al., 2023b)	2023	context	trajectory	descriptor of execution...		P	MT-GRPO Iterative... (Modersena et al., 2026)	2026	context	interaction	interaction		A
Reflection (Shim et al., 2023)	2023	ext. memory	trajectory	verbal self-reflection after...		P	Self-Guide internal... (Wang et al., 2026e)	2026	written	trajectory	self-generated guidance as...	o	A
RCI (Kim et al., 2023)	2023	context	interaction	recursive self-criticism of...		P	SAVER (Yuan et al., 2026)	2026	context	none	adversarial audit localizing...	o	A
Self-Refine (Madala et al., 2023)	2023	context	output	self-generated...		P	ReflectChain... (Lao, 2026a)	2026	world model	trajectory	latent trajectory rehearsal...	o	A
Generative Agents (Park et al., 2023)	2023	ext. memory	interaction	observation stream, periodic...		P	Cyclic subtask graphs (Charzedeine and Saab, 2026)	2026	written	none	state-analysis-and-routing...	o	A
Self-Debug (Chen et al., 2023b)	2023	context	output	execution feedback +...		P	CAPO (Wang et al., 2026a)	2026	context	interaction	none		A
Tree of Thoughts (Yao et al., 2023a)	2023	context	trajectory	self-evaluation of...		P	AEM adaptive entropy... (Zhao et al., 2026a)	2026	context	trajectory	response-level surprisal x...	o	A
MemoryBank (Zhong et al., 2023)	2023	ext. memory	output	Ebbinghaus-style forgetting...		P	T2PO (Wang et al., 2026b)	2026	context	interaction	token-level...	o	A
RecurrentGPT (Zhou et al., 2023c)	2023	written	output	LLM rewrites its own...		A	MedExAgent (Gao et al., 2026a)	2026	context	output	none		A
LLM-MCTS (Zhou et al., 2023b)	2023	world model	trajectory	LLM-as-world-model state...	o	P	TRACE turn credit for... (He et al., 2026b)	2026	context	interaction	leave-one-turn-out semantic...	o	A
RAP (Hao et al., 2023)	2023	world model	trajectory	LLM-as-world-model predicted...		P	PCA post-based credit (Liu et al., 2026a)	2026	context	trajectory	prior-step potential-based...	o	A
Voyager (Wang et al., 2023a)	2023	ext. memory	trajectory	environment execution...		P	CHAL (Giovannelli and Kent, 2026)	2026	written	trajectory	gradient-informed revision on...	o	A
Expel. (Zhao et al., 2023a)	2023	ext. memory	trajectory	cross-task insight extraction...		P	Why world models (Alawad et al., 2026)	2026	context	none	none		A
CoALA (Sumers et al., 2023)	2023	ext. memory	none	conceptual		P	Predictor (Six et al., 2026)	2026	world model	world-model loss	belief-based likelihood score...	o	A
TradingGPT (Li et al., 2023)	2023	ext. memory	interaction	layered memory decay +...		A	Belief Engine stance... (Yang et al., 2026b)	2026	prob. memory	none	log-odds update with evidence...	✓	A
FaR / T4D (Zhou et al., 2023b)	2023	context	interaction	foresee-and-reflect...		P	CAGE-2 compound agent... (Bogdanov et al., 2026)	2026	ext. memory	none	deterministic state-tracking...	o	A
LATS (Zhou et al., 2023a)	2023	ext. memory	trajectory	MCTS backup of LM-valued...	o	P	Representation... (Xue, 2026)	2026	context	output	structured risk feedback	o	A
FinAct (Chen et al., 2023a)	2023	context	trajectory	supervised fine-tuning on...		A	CM1 (Srivastava, 2026)	2026	ext. memory	none	counterfactual answer-score...	o	A
MemGPT (Packer et al., 2023)	2023	ext. memory	output	self-directed memory paging...		A	SERL selective... (Li et al., 2026b)	2026	context	interaction	per-step environment feedback...	o	A
AgentTuning (Zeng et al., 2023)	2023	context	trajectory	hybrid instruction tuning on...		P	Memory-R2 / LoGo-GRPO (Yan et al., 2026a)	2026	ext. memory	none	local rollouts from shared...	o	A
FinMem (Yu et al., 2023)	2023	ext. memory	interaction	layered memory with...		P	SkiliC (Lin et al., 2026a)	2026	context	trajectory	skill-vs-no-skill reward gap	o	A
Reflect-RL (Zhou et al., 2024a)	2024	written	interaction	fraction reflection model...		P	AIK financial harness (Borjigin et al., 2026)	2026	ext. memory	none	passive knowledge injection...	o	A
FinAgent (Zhang et al., 2024a)	2024	ext. memory	interaction	low-level and high-level...		P	TOKI (Wang, 2026)	2026	ext. memory	none	binomial operators	o	A
AgentGym (Xi et al., 2024)	2024	context	trajectory	AgentEvolve		A	Q-Evolve (Zhang et al., 2026e)	2026	context	trajectory	advantage from...	o	A
Hypothetical Minds (Cross et al., 2024)	2024	written	none	hypotheses about other...		A	HiPF (Diao et al., 2026)	2026	written	trajectory	subgoal completion +...	o	A
FinCon (Yu et al., 2024)	2024	ext. memory	none	episodic self-critique...		P	ProPlay (Ma et al., 2026)	2026	world model	trajectory	post-episode graph refinement...	o	A
Agent Q (Putta et al., 2024)	2024	ext. memory	trajectory	MCTS with AI self-critique...		A	POMDP validation... (Dixon, 2026c)	2026	ext. filter	none	Bayesian	✓	A
Agent Workflow Memory (Wang et al., 2024)	2024	ext. memory	none	induction of reusable...		A	G2PO (Wang et al., 2026f)	2026	context	trajectory	foreight-conditioned TD...	o	A
Talker Reasoner (Christakopoulos and Mourad, 2024)	2024	written	none	Reasoner writes/updates...		A	Internalizing the Future (Zhang et al., 2026d)	2026	written	trajectory	foreight-conditioned RL...	o	A
QUBE (Kim et al., 2024)	2024	written	none	question-answering over...		P	Hybrid-WM (Song and Cai, 2026a)	2026	world model	world-model loss	consistency gate	✓	A
TradingAgents (Xiao et al., 2024)	2024	context	output	multi-agent debate and...		A	Governance-aware POMDP... (Dixon, 2026a)	2026	ext. filter	none	Bayesian update on evidence...	o	A
AgentRefine (Fu et al., 2025)	2025	context	trajectory	environment observation...		P	BayoEvolve (Wu et al., 2026b)	2026	ext. filter	none	posterior refit after each...	✓	A
LOOP (Chen et al., 2025a)	2025	context	trajectory	environment feedback via API...		P	WorldEvolver (Zhang et al., 2026c)	2026	world model	world-model loss	prediction-observation...	o	A
A-Mem (Xu et al., 2025b)	2025	ext. memory	output	memory evolution		P	WM-SAR (Song and Cai, 2026b)	2026	world model	trajectory	spectral error amplification	o	A
ReSearch (Chen et al., 2025b)	2025	context	output	retrieved documents...		A	STAPO... (Qi et al., 2026)	2026	context	trajectory	normalized-entropy outlier...	o	A
RAGEN / StarPO (Wang et al., 2025)	2025	context	trajectory	search results inserted into...		P	Auditing... (Gao et al., 2026b)	2026	ext. memory	none	logged per-turn belief updates	✓	A
Mem0 (Chikara et al., 2025)	2025	ext. memory	none	environment reward...		A	RLAW (Ali and Tseng, 2026)	2026	ext. memory	interaction	critique score	o	A
TTI (Shen et al., 2025)	2025	context	trajectory	LLM-driven extraction...		A	NeSyFS (Xu and Fekri, 2026)	2026	ext. memory	interaction	repeated fast-action failure...	o	A
MEMI (Zhou et al., 2025c)	2025	written	outcome (written state)	additional environment...		A	MADE (Liu et al., 2026b)	2026	ext. memory	interaction	execution failure ->...	o	A
MemAgent (Yu et al., 2025)	2025	written	outcome (written state)	chunk-wise overwrite of...		P	TPCO turn-level credit (Liao et al., 2026b)	2026	context	interaction	majority vote over arithmetic	o	A
ARPO (Dong et al., 2025a)	2025	context	trajectory	token entropy spike after...		A	When Memory Updates but... (Sun and He, 2026)	2026	ext. memory	none	state dependency repair	o	A
Memorito (Zhou et al., 2025b)	2025	ext. memory	trajectory	environment feedback, iter...		P	LAPO (Ren et al., 2026)	2026	ext. memory	none	learned...	o	A
Agent-R1 (Cheng et al., 2025)	2025	context	trajectory	tool/environment feedback per...		A	GPm (Xu, 2026)	2026	ext. memory	none	symbolic check	o	A
MATTRL test-time... (Hu et al., 2026)	2026	ext. memory	interaction	turn-level credit for...		A	CEST (Wang et al., 2026g)	2026	context	interaction	verified turn reward with...	o	A
SuS strategy-aware... (Kashirsky and Makarov, 2026)	2026	context	output	surprise reduction...		A	HIDITTR (Guo et al., 2026)	2026	context	interaction	group tool-call success rate	o	A
External affective... (Sobharwal, 2026)	2026	ext. memory	none	exponential smoothing or...		A	Intentional	2026	context	interaction	intentional	o	A
PCE (Seo et al., 2026)	2026	context	none	scenario likelihood scoring...		A	BCO (Chen et al., 2026b)	2026	written	none	evaluation of new candidate...	✓	A
HIFER (Peng et al., 2026)	2026	context	trajectory	self-supervised world-model...		A	EvoSCAM capture belief... (Zhao et al., 2026b)	2026	written	none	prediction-observation...	✓	A
RB-VLA (Bagaria et al., 2026)	2026	world model	world-model loss	self-supervised world-model...	o	A	CAPTURE (Hossain et al., 2026)	2026	world model	world-model loss	uncertainty-triggered...	o	A
REDDER (Hossain et al., 2026)	2026	prob. memory	interaction	judge feedback updates...		A	PPPO (Zhang et al., 2026)	2026	context	interaction	potential difference between...	o	A
BrainSolveFlow (Yan et al., 2026)	2026	ext. memory	interaction	episodic entropy over samples...		A	Semantic Bayesian World Model (Soru, 2026)	2026	ext. filter	none	Semantic Bayesian World Model...	o	A
MICA (Zhang et al., 2026b)	2026	ext. memory	interaction	none		A	Belief-State Engine (Chattapadhyay and Halker, 2026)	2026	ext. filter	none (prompting)	Bayesian posterior	o	A

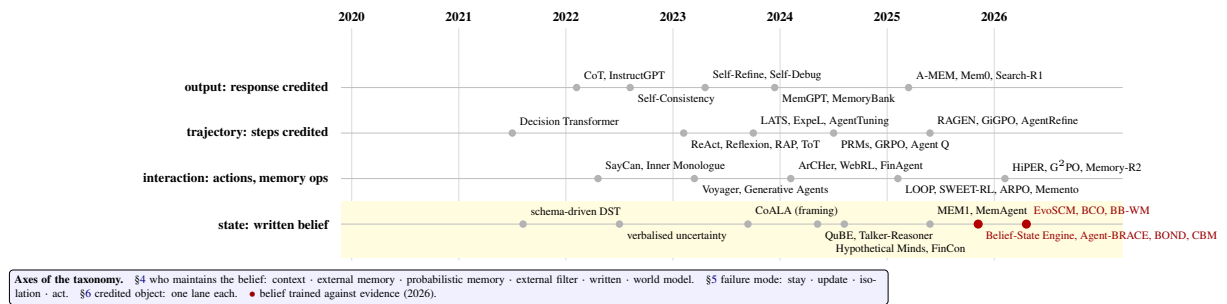


Figure 7: Where credit has gone, 2020–2026. Lanes are credited objects; Table 12 lists every screened system. Written beliefs exist from dialogue state tracking onward, but until 2026 they are credited by task outcome or not trained at all; the red points are the first to give the belief its own learning signal.